Stone Conservation Project

Monica Bahamondez Prieto, is in charge of the island's restoration project under the Dirección de Bibliotecas, Archivos y Museos, Santiago de Chile. Her report, in Spanish, is followed by an English translation.

by Monica Bahamondez Prieto

Para nadie es un secreto que estado de deterioro en que se encuentra gran cantidad de moai de Isla de Pascua es de carácter grave. Por esta razón se elaboró un proyecto de conservación del material litico de la isla, el que se enmarca dentro del Proyecto CHI/79/013, suscrito entre el PNUD y la Dirección de Bibliotecas, Archivos y Museos. Comienza en el año 80 con traida a la isla del experto polaco Sr. W. Domaslowski, quien propone en primera instancia un método de Conservación de la Piedra. Posteriormente se envio a la Sra. Mónica Bahamondez, Ingeniero Quimico, a realizar un curso de especialización en Conservación de Piedra (Venecia 83), con el fin de analizar la factibilidad de aplicar el método propuesto y tomar bajo su cargo la etapa experimental.

La empresa alemana Wacker, fabricante de los productos recomendados prestó la asesoria del Prof. Roth, Quimico de dicha empresa, quien supervisó la aplicación de las resinas.

El trabajo es de carácter experimental y consiste básicamente en consolidar la piedra para devolverle su durenza original y posteriormente hidrofobizarle, es decir, impedir la penetración de agua, principal causante del deterioro. Los resultados obervados a la fecha son altamente positivos con lo que esperamos retardar al máximo el deterioro de los moais. En el presente año se ha planificado la intervención en 3 esculturas ubicadas en distintas zonas de la isla lo que permitirá una buena evaluación de la etapa experimental.

[It is no secret that the state of deterioration of a great number of moai on Easter Island is very grave. For this reason, a stone conservation project was developed as part of Project CHI/79/013, co-sponsored by the PNUD and the management of Libraries, Archives and Museums. It began in 1980, when the Polish expert, W. Domaslowski, was brought to the island. He first proposed a method of stone conservation. Later, Sra. Bahamondez, a Chemical Engineer, was sent to present a special course in stone conservation for the purpose of analyzing the feasibility of using the proposed method and to take charge of the experimental stage.

The German firm Waker, maker of the recommended products, provided the consulting services of professor Roth, chemist at the firm, who supervised the application of the resins.

The work is experimental and consists basically of consolidating the stone in order to return it to the original hardness and then prevent the penetration of water, which is the principle cause of deterioration. The observed results to date are very positive and it is hoped that this process will retard the deterioration of the moai. This year three sculptures will be treated in different parts of the island; this will enable us to make an evaluation of the experimental process.]

Report on Recent Archaeology on Rapa Nui

by Dr. Christopher Stevenson

As part of a long term research project concerned with the study of prehistoric Easter Island economy and socio-political structure, excavations were conducted at Site 6-356 with the assistance of Earthwatch volunteers under the direction of Dr. Christopher Stevenson and Ms. Felicia Rounds. The excavations entailed the intensive excavation of a single open-air residential site located in the southwestern portion of the island. This undertaking represents the first thorough investigation of a major habitation site ever conducted on Easter Island.

Research objectives for the first field season focused on the identification of site workshop areas and the investigation of specific cultural features. In order to realize these objectives, a complete surface collection of the artifactual material from the surface of the site was taken. The entire site area was divided into 192 five meter square units and all artifacts of obsidian and basalt were collected. A preliminary mapping of the absolute numbers of artifacts indicated that well defined areas of high artifact concentration were present across the site area. These zones of high density appear to represent areas where obsidian was worked into finished tools. Five distinct concentrations were identified.

After the surface collection, the entire site was tested with one meter square excavation units positioned at systematic intervals across the site. An additional sample of the stone tool assemblage was collected and part of a purposefully buries human skull was recovered. Also encountered were five basalt hoes. To our knowledge, this is the first reported discovery of this tool type from the island.

Architectural features at the site consisted of a subterranean *manavai* (garden enclosure), a *hare moa* (stone chicken coop), an *ana kionga* (refuge cave), the remains of a house entrance feature, and a deep cooking midden with *umu* (earth oven) preserved on the surface. Test excavations with the midden deposits indicated that cultural material extended within well defined stratigraphy to a depth of over one meter. Ample charcoal from these deposits was recovered for radiocarbon dating.

The excavation of the *ana kionga* provided a rich faunal data set dating to the post 1680 AD period. A study of the cave interior and surrounding deposits indicated that the cave had been enlarged by prehistoric excavation. After this expansion, a small interior room was walled off with stones and an entrance tunnel was created to restrict access to the cavity. Debris from the enlargement of the cave was dumped outside the cave and was used to conceal the basalt outcrop under which the cave was located and a petroglyph of pecked lines cut into the same outcrop. All indications suggest that the cave was purposely fortifiedand camoflaged in order to prevent identification by hostile persons.

Twelve test units were placed inside the cave and produced thousands of fish, rat, and chicken bones from the 3-

continued on page 6.

Fonck Easter Island Collection

by José Miguel Ramirez

The Fonck Society Museum is a small but important institution located in the tourist capital of Chile, Viña del Mar, two hours from Santiago.

Founded in 1937, it has played an important role in the archaeological investigations of the central coast of Chile.

Thanks to many collectors, the most important section of the Museum has come from Easter Island beginning 50 years ago. Fritz Felbermayer, who founded "The Society of Friends of Easter Island," was the principle donor; he contributed 589 archaeological and ethnological objects including some of the most important and unique pieces.

The most unique article in the entire collection is a large tapa robe *(nua mahute)* 180 x 140 cm, dyed a yellowish color with 15 *manutara* (terns) and one *tangata manu* (birdman). It most probably belong to the last chief of the island. The cloak is formed of two or three layers of *mahute* (paper mulberry) sheets sewn together with fine vegetal cords, each measuring 4 cm wide, like back stitching.

In 1951 the Fonck Society brought to Viña del Mar a three meter tall *moai* from Ahu One Mahiki on the south coast of Rapa Nui.

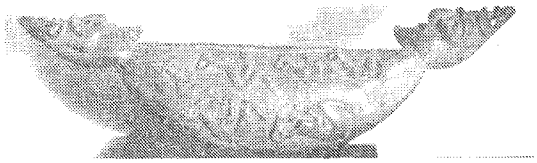
Today the Easter Island collection includes more than 1400 pieces from every period of the island's history.

Among the more exceptional pieces: a large polished adze *(toki)* 37.5 cm across, and weighing 4.8 kg; a beautiful old *reimiro*, 80 cm (another interesting *reimiro* carved with six *tangata manu* in bas relief, was stolen from a private exhibition hall following the 1985 earthquake which destroyed the old museum); a 152 cm dance paddle *(ao)*; a 118 cm decorated club *(ua)*; 8 exquisite wooden *moai tangata* (male figures); one old *moai pa'apa'a* (literally, sterile female figure); and one *moai vie* (pregnant female figure).

Also there are tattooing combs *(uhi)*; 8 incised stone pillows *(ngarua)*; 2 knives *(hoe)*; 2 axes *(ohio)*; 91 adzes *(toki)* and chisels *(kautoki)*; one large obsidian *toki*; 540 obsidian spear points *(mata'a)*; 4 knife handles, drills and other artifacts; 3 one-piece stone fishhooks *(mangai ivi)*, 5 pieces in the process of manufacture; one stone point of a composite fishhook; one composite bone fishhook *(mangai veri veri)*; 15 one-piece bone fishhooks *(mangai ivi)*; and many other bon artifacts. We have one stone anchor *(aka)*; 5 grooved sinkers 16 bone needles; one whetstone with parallel groves in on side; one slab fragment *(paenga)* with 2 bi-conical holes in th top; and one rubbing stone.

Additional pieces include various good luck objects: fer tility stones for chickens; a boulder with three finned vulva phallic stones; and others. There are 18 wood and 4 stone ba neck pendants *(tahonga)*; several old stone busts, heads an zoomorphic figures in different styles and materials; an many wooden and stone figures from modern periods; orna ments, shell necklaces, feather skirts and circlets, fibe purses, carpets and hats, and a large quantity of lava figure made in the middle 1950's. Among the interesting moder pieces are diving glasses made of wood and flint glas (collected by Felbermayer in the 1940's.) Moreover, ou library has a varied collection of books, articles, document a number of *kai kai* figures, and drawings and photograph from Easter Island. We have no *rongorongo* tablets nor o *moai kavakava*.

For the next number of CLAVA, the scientific magazine the Museum, we are collecting original contributions abo Easter Island archaeology and anthropology. It will be edite in September 1988.



Editor's Note: The Fonck Society Museum has one of the finest off-island collections of Rapa Nui artifacts in the world. I *most urgently needs* funds for research, preservation and fo exhibits. PLEASE HELP! The Museum needs your support.

Readers can become a Cooperating Associate Member (U.S $60 annually) and obtain all the various Museum Publications Also, the Catalog of the Rapa Nui collection is available for U.S $10. Checks should be to the order of "Sociedad de Arqueologi Dr. Fonck" and sent to Casilla 951, Viña del Mar, Chile.

...Recent Archaeology *continued....*

cm of cultural deposits that had accumulated. A single European glass trade bead and several very fine examples of bone needles and awls were also recovered. The glass bead also indicated that the cave was occupied after European contact (post 1722 AD) and possibly until around 1850 AD. Obsidian hydration and radiocarbon dating will be used to provide additional chronometric assays for the deposits in order to better define the period of cave construction and use. Even without these determinations, it is clear that the cave deposits represent a very limited period of time and provide an assemblage of food remains possibly for the later period of island prehistory.

The excavation of Easter Island residential sites will continue again next year. Persons interested in participating this project should contact Jay Kolin at Earthwatch Expe tions, Watertown, Mass 02172; phone (617) 926-8200.

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Center for South Pacific Studies

A Center for South Pacific Studies at the University of New South Wales, Australia, has been organized to research and distribute information on the peoples and places of the South Pacific. For further details about the Center, contact Dr. Gran McCall, School of Sociology, The University of NSW, Kensington, New South Wales 2033, Australia.